ACTL3142 Tutorial

Week 4: Logistic & Poisson Regression

Nick Guo

School of Risk and Actuarial Studies, UNSW Sydney

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Recap from Last Week

From Weeks 2–3 (Linear Regression):

- Model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$
- Fit by OLS (minimise RSS): $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- β_j: partial effect of X_j on Y, holding the other predictors fixed
- Inference via t-tests / F-test; fit measured by R^2

New problem: linear regression assumes Y is continuous and that its mean is linear over all of $\mathbb{R}$. What if

- Y is **binary** (default / no default, claim / no claim)?
- Y is a **count** (number of claims, number of deaths)?

The linear model then gives nonsensical predictions (probabilities outside $[0, 1]$, negative or non-integer counts).

Today: **logistic regression**, **evaluating classifiers**, and **Poisson regression**, all unified as **GLMs**.

Why Standard Linear Regression Fails for Classification

Suppose $Y \in \{0, 1\}$ (say $1 =$ default). We want the conditional probability

$$p(X) := P(Y = 1 \mid X) \qquad \text{(the same } p(X) \text{ notation as ISLR).}$$

Naive idea (linear probability model): regress the $0/1$ response directly,

$$p(X) \approx \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

Problems:

- The right-hand side ranges over all of $\mathbb{R}$, so fitted "probabilities" can be < 0 or > 1
- The implied error variance is not constant

The fix

Don't model the probability linearly. First transform it onto a scale that is **unbounded**, then put the linear predictor there. That transform is the **logit**.

Building Block: Odds and Log-Odds

Start from a probability $p = P(Y = 1) \in [0, 1]$, which is awkward to model linearly (it is bounded).

- **Odds:** $\dfrac{p}{1-p} \in [0, \infty)$, "how many times more likely than not"
- **Log-odds (logit):** $\text{logit}(p) = \ln \dfrac{p}{1-p} \in (-\infty, \infty)$, now *unbounded*

p	odds $= \frac{p}{1-p}$	logit $= \ln \frac{p}{1-p}$
0.10	0.11	-2.20
0.50	1.00	0.00
0.75	3.00	1.10
0.90	9.00	2.20

Why the logit?

Probabilities live in $[0, 1]$, but a linear predictor lives in $\mathbb{R}$. The logit stretches $[0, 1]$ out to the whole real line, so a linear model becomes sensible.

The Logistic Regression Model

Recall $p(X) = P(Y = 1 \mid X)$. Model its **log-odds** as linear in the predictors:

$$\ln\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p =: \eta.$$

Invert to recover the probability (the **logistic / sigmoid function**):

$$p(X) = \frac{e^{\eta}}{1 + e^{\eta}} = \frac{1}{1 + e^{-\eta}} \in (0, 1).$$

Equivalently, the **odds** are multiplicative in the predictors:

$$\frac{p(X)}{1 - p(X)} = e^{\eta} = e^{\beta_0}\, e^{\beta_1 X_1} \cdots e^{\beta_p X_p}.$$

The S-shaped sigmoid squashes any real η into $(0, 1)$, so predictions are always valid probabilities, whatever the predictor values.

Interpreting the Coefficients

A one-unit increase in X_j (others held fixed):

- **Log-odds scale:** adds β_j (additive, like OLS but on log-odds)
- **Odds scale:** multiplies the odds by e^{β_j}
  - $\beta_j > 0 \Rightarrow e^{\beta_j} > 1$: odds (and probability) increase
  - $\beta_j < 0 \Rightarrow e^{\beta_j} < 1$: odds decrease
- **Probability scale:** the change is *not* constant. The sigmoid is steep near $p = 0.5$ and flat near 0 or 1, so the same β_j moves p a lot in the middle and barely at the extremes

Multiplicative, not additive

In MLR, β_j is a constant *additive* effect on $\mathrm{E}[Y]$. In logistic regression, β_j is a constant additive effect on the *log-odds*, i.e. a constant *multiplicative* effect (e^{β_j}) on the odds, but a *varying* effect on the probability.

Q1: Probability of an A

⋆ Solution (ISLR2, Q4.6) Suppose we collect data for a group of students in a statistics class with variables X_1 =hours studied, X_2 = undergrad GPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$, $\hat{\beta}_2 = 1$.

1. a. Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

   b. How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

Estimation by Maximum Likelihood

There is no closed form like OLS. We choose $\boldsymbol{\beta}$ to maximise the likelihood of the observed $0/1$ labels.

For observation i, write the linear predictor $\eta_i = \mathbf{x}_i\boldsymbol{\beta}$ and the fitted probability

$$p_i \;=\; p(\mathbf{x}_i) \;=\; P(Y_i = 1 \mid \mathbf{x}_i) \;=\; \frac{1}{1 + e^{-\eta_i}}.$$

Each $Y_i \sim \text{Bernoulli}(p_i)$, so the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big[\, y_i \ln p_i + (1 - y_i) \ln(1 - p_i) \Big].$$

Maximise **numerically** (no $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ shortcut); software uses Newton's method / iteratively reweighted least squares.

There is no exact R^2 here (no $TSS = MSS + RSS$ split). Compare models with **AIC** *or deviance, and test single coefficients with the Wald z-statistic (R's `z value`, not a t) in the summary output. Collinearity still inflates standard errors, just as in MLR.*

From a Probability to a Classification

The model outputs a probability $\hat{p}(X)$. To classify, apply a **threshold** c:

$$\hat{Y} = \begin{cases} 1 & \hat{p}(X) > c \\ 0 & \hat{p}(X) \leq c \end{cases}$$

- Default $c = 0.5$ (assign to the more likely class), but c is a **choice**
- Lowering c catches more positives (fewer false negatives) at the cost of more false positives, and vice versa

The "right" threshold depends on the relative cost of the two error types. In fraud detection or disease screening, missing a positive (false negative) is far worse than a false alarm, so we lower c. The ROC curve (later) shows this trade-off across all c.

The KNN Classifier Revisited

A **non-parametric** method we will need for Q3 and the applied question. To predict at a new point $\mathbf{x}_0$, find the K training points closest to $\mathbf{x}_0$, then:

- **Classification:** predict the **majority class** among the K neighbours
- **Regression:** predict the **average response** among the K neighbours

K controls flexibility (the bias-variance trade-off):

- $K = 1$: classify by the single nearest neighbour. Fits the training data perfectly (**0% training error**) but a very wiggly boundary, high variance, overfits
- Large K: smoother boundary, more bias, lower variance

Because 1-NN always has 0% training error, training error tells you nothing about how it generalises. Always judge KNN on held-out data, this is the key to Q3.

Q3: Logistic Regression vs 1-NN

3. ⋆ Solution (ISLR2, Q4.8) Suppose that we take a data set, divide it into equally-sized training and test sets, and then try out two different classification procedures. First we use logistic regression and get an error rate of 20% on the training data and 30% on the test data. Next we use 1-nearest neighbors (i.e. $K = 1$) and get an average error rate (averaged over both test and training data sets) of 18%. Based on these results, which method should we prefer to use for classification of new observations? Why?

Evaluating Classifiers I: From RSS to the Confusion Matrix

In regression we judged fit with RSS / MSE: squared distance from a *continuous* target. That does not transfer to classification:

- a class label is not a magnitude (the 0/1 coding is arbitrary), so "squared distance" is not meaningful
- we care about the *type* of mistake (FP vs FN), which a single number cannot express

So we **count** predictions against the truth instead. Accuracy = fraction correct, error rate $= \frac{1}{n}\sum_i I(y_i \neq \hat{y}_i)$.

Two error types: **FP** (predict 1, truth 0; Type I) and **FN** (predict 0, truth 1; Type II). Cross-tabulate them, the **confusion matrix**:

	Actual 0	**Actual 1**
Predicted 0	TN	FN
Predicted 1	FP	TP

Evaluating Classifiers II: Recall, Specificity, Precision

Accuracy alone can mislead (think imbalanced classes), so look at each class separately:

- **Recall** (sensitivity, TPR) $= \dfrac{TP}{TP + FN}$: of the actual positives, how many did we catch?
- **Specificity** (TNR) $= \dfrac{TN}{TN + FP}$: of the actual negatives, how many did we correctly clear?
- **Precision** $= \dfrac{TP}{TP + FP}$: of those we flagged positive, how many really are?

Accuracy can lie (cheating detector)

A detector flags 30 students, only 9 of whom actually cheated: accuracy 97.8%, specificity 97.9%, but **precision just** 30%. With rare positives, strong accuracy and specificity can hide terrible precision.

ROC Curve and AUC

The threshold c trades recall against false positives. Sweep c from 1 down to 0 and plot:

- y-axis: True Positive Rate = recall
- x-axis: False Positive Rate $= 1 -$ specificity

The traced-out curve is the **ROC curve**.

- Top-left corner = perfect (catch all positives, no false alarms)
- Diagonal = random guessing

AUC (area under the curve) $\in [0, 1]$ summarises performance across *all* thresholds:

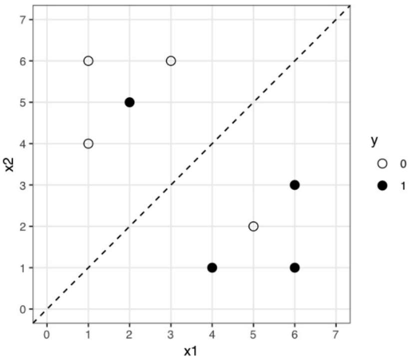
$$\text{AUC} = 0.5 \;\Rightarrow\; \text{chance}, \qquad \text{AUC} = 1 \;\Rightarrow\; \text{perfect}, \qquad \text{AUC} > 0.7 \;\Rightarrow\; \text{re}$$

AUC is threshold-free, so it is a good single number for comparing models before you commit to an operating threshold. We sketch one by hand in Q4(c).

Q4: Decision Boundary, Confusion Matrix, ROC

⋆ Solution Consider a classification problem with two continuous inputs, x_1 and x_2, and a binary (0/1) target variable, y. There are 8 training cases, plotted in the following figure.

4.



Cases where $y = 1$ (positive cases) are plotted as black dots and cases where $y = 0$ (negative cases) as white dots, with the location of the dot giving the inputs, x_1 and x_2, for that training case. A logistic regression has been fitted to these data with estimated regression equation:

$$\log\left(\frac{\mathbb{P}(Y = 1|x_1, x_2)}{1 - \mathbb{P}(Y = 1|x_1, x_2)}\right) = x_1 - x_2.$$

The corresponding classification decision boundary using a threshold of 0.5 probability is given by $x_2 = x_1$ and is represented by the dashed line in the figure (an observation is assigned to class $y = 1$ if $\mathbb{P}(Y = 1|x_1, x_2) > 0.5$).

Applied Q1: Weekly Data (parts a–c)

1. ⋆ Solution (ISLR2, Q4.13) This question should be answered using the `Weekly` data set, which is part of the `ISLR2` package. This data is similar in nature to the `Smarket` data from this chapter's lab, except that it contains 1,089 weekly returns for 21 years, from the beginning of 1990 to the end of 2010.

   a. Produce some numerical and graphical summaries of the `Weekly` data. Do there appear to be any patterns?

   b. Use the full data set to perform a logistic regression with `Direction` as the response and the five lag variables plus `Volume` as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

   c. Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

Applied Q1: Weekly Data (parts d–g)

d. Now fit the logistic regression model using a training data period from 1990 to 2008, with `Lag2` as the only predictor. Compute the confusion matrix and the overall fraction of correct predictions for the held out data (that is, the data from 2009 and 2010).

e. Repeat (d) using KNN with $K = 1$.

f. Which of these methods appears to provide the best results on this data?

g. Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.

Applied Q1: Hints

Pull this question up on your side. Key pointers for each part:

(a) `library(ISLR2)`, then `summary(Weekly)` and `pairs()` / `cor(Weekly[,-9])`. Volume trends up over time; the lag variables show little obvious pattern.

(b) `glm(Direction ~ Lag1+...+Lag5+Volume, data=Weekly, family=binomial)`, then `summary()`. Only **Lag2** is significant ($p \approx 0.03$).

(c) `predict(fit, type="response")`, classify at 0.5, `table(pred, Direction)`. Accuracy $\approx 56\%$, but it predicts "Up" almost always: high recall on Up, very poor on Down.

(d) Train on `Year <= 2008`, test on 2009–2010, **Lag2 only**. Test accuracy $\approx 62.5\%$.

(e) `library(class)`; `knn(train.X, test.X, train.Dir, k=1)`. Accuracy $\approx 50\%$ (no better than chance).

(f) Logistic (Lag2) beats 1-NN on the held-out data.

(g) Experiment with transformations, interactions, and larger K (standardise predictors when using several, since KNN uses distances); report the best held-out confusion matrix you find.

Poisson Regression: Modelling Counts

Now $Y \in \{0, 1, 2, \dots\}$ is a **count**: number of claims on a policy, number of deaths in an age band.

Why not linear regression? Predictions can go negative, the variance is assumed constant, and counts are integers.
Why not $\log Y$? $\log 0$ is undefined, and zero counts are common.

Model: $Y \mid X \sim \text{Poisson}(\lambda)$ with the **log link**:

$$\ln \lambda(X) = \beta_0+\beta_1 X_1+\cdots+\beta_p X_p \iff \mathrm{E}[Y \mid X] = \lambda(X) = e^{\beta_0+\beta_1 X_1+\cdots+}$$

The exponential guarantees a positive mean, exactly as the sigmoid guaranteed a valid probability. Same idea, different link.

Poisson Regression: Interpretation and Properties

- **Coefficients:** a one-unit increase in X_j multiplies $\mathrm{E}[Y]$ by e^{β_j} (multiplicative, just like the odds in logistic)
- **Key property:** $\mathrm{E}[Y] = \mathrm{Var}(Y) = \lambda$. The variance grows with the mean, it is *not* constant
- **Estimation:** by MLE; fitted values $\hat{\lambda}_i$ are always positive

Mean = variance

The single parameter λ controls both the mean and the variance. If the data show $\mathrm{Var}(Y) > \mathrm{E}[Y]$ ("overdispersion"), plain Poisson understates uncertainty, and you would reach for quasi-Poisson or negative binomial (beyond this course).

As with logistic, there is no exact R^2; compare models with AIC or deviance.

The Bigger Picture: Generalised Linear Models

Logistic and Poisson are not one-off tricks. Both are **GLMs**: keep a linear predictor $\eta = \mathbf{X}\boldsymbol{\beta}$, but

1. pick a **distribution** for Y that suits the data, and
2. pick a **link** g so that $g(\mathsf{E}[Y]) = \eta$ maps the mean onto $\mathbb{R}$.

Model	**Response**	**Distribution**	**Link** $g(\mu)$	$\mathsf{E}[Y \mid X]$
Linear	continuous	Normal	μ	$\mathbf{X}\boldsymbol{\beta}$
Logistic	binary	Bernoulli	$\ln \frac{\mu}{1-\mu}$	$\dfrac{1}{1+e^{-\mathbf{X}\boldsymbol{\beta}}}$
Poisson	count	Poisson	$\ln \mu$	$e^{\mathbf{X}\boldsymbol{\beta}}$

All fitted by MLE; all share the same modelling caveats (collinearity, predictor selection via AIC, etc.).

This is the unifying idea of the week. Once you see linear regression as "identity link + Normal", logistic and Poisson are just different (link, distribution) choices for the same machinery.

Thanks for coming! See you next week.